

Lekkala Ganesh

India | +91-8688479455 | lekkalaganesh14@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

High-velocity AI Application Engineer & Full-Stack Architect with a proven track record of shipping 9+ production-ready applications within 3 months. Specialized in Generative AI orchestration (RAG, Diffusion Models) and high-compute VRAM optimization, reducing GPU operational costs by 75%. Expert at leveraging Agentic AI workflows to accelerate development cycles by 3x while maintaining zero-defect production standards. Seeking elite software engineering roles in Bangalore, Hyderabad, or Chennai tech hubs.

TECHNICAL CORE (2026 SPECIALIZATION)

AI Orchestration & LLMOps: PyTorch, Hugging Face (Diffusers, Transformers), RAG Pipelines, 4-bit Quantization (bitsandbytes), CUDA, Multi-agent Workflows, Vector Databases (Pinecone, ChromaDB), Model Fallback Architectures

Platform & Architecture: Distributed Systems, System Design, Microservices, REST/WebSocket APIs, PostgreSQL (Row-Level Security), Supabase, MongoDB Atlas

Product Delivery: React, Next.js (SSR/SSG), TypeScript, Node.js, FastAPI, Tailwind CSS, Framer Motion, Three.js

Cloud & DevOps (GitOps): Docker, Kubernetes (K8s), GitHub Actions (CI/CD), Terraform (IaC), Linux Shell Scripting, Vercel, Prometheus/Grafana

Quality & Compliance: Vitest, Playwright (E2E), WCAG 2.1 AA Accessibility, SEO (JSON-LD), HIPAA-aligned Data Handling

PROFESSIONAL EXPERIENCE

Jr. Software Developer (Full-Stack Lead)

AlterSquare

Remote

Dec 2025 – Present

- **Founding Engineer Ownership:** Owned the end-to-end SDLC for 9 client-facing web applications; delivered all projects from requirement gathering to production within initial 90 days with zero critical bugs reported post-deployment.
- **AI System Integration:** Engineered a multi-provider AI fallback system (GPT-4o, Gemini, OpenRouter) to ensure 99.9% uptime for mission-critical apps, including real-time SSE voice streaming ([VoxAI](#)) and 3D shadow simulations ([SunScope Pro](#)).
- **Security & Scalability:** Architected high-security backends using Row-Level Security (RLS) and JWT authentication on Supabase/PostgreSQL, managing secure data for financial ([Finance Tracker](#)) and contractor platforms ([2A Construction](#)).
- **Accessibility Excellence:** Achieved 100% WCAG 2.1 AA compliance across all products using custom ally hooks and automated Playwright testing, significantly expanding the accessible user base for client startups.

Web Developer Intern

Dash Technologies Inc.

Ahmedabad, Gujarat

Jan 2025 – Jul 2025

- **Healthcare UI Delivery:** Built 12+ modular, responsive React components for a patient-facing portal, reducing accessibility audit findings by 40% through semantic HTML and HIPAA-aligned data handling.
- **Cross-Browser Engineering:** Ensured 100% visual parity across Chrome, Firefox, Safari, and Edge for complex modules like appointment scheduling and patient history.

KEY TECHNICAL PROJECTS

AuraGen – High-Compute AI Generation Platform | *PyTorch, HF Diffusers, FastAPI, Next.js, CUDA*

[GitHub](#)

- **VRAM Optimization:** Engineered a specialized 4GB VRAM optimization stack (NF4 quantization, attention slicing, VAE tiling) that reduced model memory requirements from 12.8GB to 3.2GB, enabling text-to-video (Wan 2.1) on consumer GPUs.
- **System Orchestration:** Designed a real-time job queue with WebSocket progress streaming and hybrid cloud burst routing (HuggingFace Inference API fallback) to handle peak inference loads.

Trip Vault – Agentic AI Travel Assistant | *GPT-4o, Supabase, Next.js, Playwright*

[Live](#)

- **Complex Workflows:** Developed a GPT-4o powered agentic system capable of executing 25+ sequential tool calls for itinerary planning, real-time API data retrieval, and automated proposal generation.

SymptomChecker AI – ML-Powered Diagnostic Engine | *Flask, Scikit-learn, React, MongoDB, JWT*

[GitHub](#)

- **Model Accuracy:** Built a Scikit-learn prediction engine achieving 95%+ accuracy across 40+ medical conditions, integrated with location-based hospital recommendation APIs.

EDUCATION

Parul University

Bachelor of Technology in Computer Science & Engineering; GPA: 7.53/10

Vadodara, Gujarat

Jul 2022 – Jun 2025